

Akshat Patil

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EDUCATION

Medhavi Skills University

B.Tech in Computer Science & Engineering (AI/ML)

Sep 2025 – Jun 2029

Bengaluru, KA

EXPERIENCE

Processing Foundation • WebGL & WebGPU

Dec 2025 – Present

- p5.js Steward on the core team and WebGPU graphics steward, reviewing contributions and guiding project direction alongside maintainers
- Engineered an instancing API (`instances(n).sphere()`) for the p5.strands WebGPU renderer, merged into core
- Redesigned GPU compute dispatch to auto-spread large 1D workloads across 2D dimensions, enabling 1M+ particle simulations
- 32 merged PRs across p5.js core, web editor and website, including GPU-to-CPU transfer via storage buffers and WebGL visual regression test infrastructure

CNCF - KubeStellar • Distributed Systems & Automation

Apr 2026 – May 2026

- Reached 3rd highest contributor by systematically auditing the console and filing 500+ verified bug reports covering i18n gaps, silent failures and broken state handling
- Resolved a race condition causing data loss across server restarts by migrating in-memory state to persistent Kubernetes ConfigMaps
- Shipped 12 merged PRs delivering monitoring presets for Harbor Registry, Knative, Fluid, in-toto and Notary

LFDT - Hiero SDK • Automation & Tooling

Apr 2026 – Present

- Built hiero-automate, a config-driven PR automation pipeline with merge-conflict detection and ESLint integration
- Shipped 15 merged PRs across the hiero-sdk-cpp and hiero-website repos, covering build tooling, CI fixes and SDK maintenance

PROJECTS

Split-it-Wise • Next.js • Firebase • Firestore

GitHub | Live

Group expense splitter used daily by hostel residents

- Built and deployed a group expense splitter that was organically adopted by roommates and floor residents for daily, real-money cost tracking
- Handled all money maths in integer paise with unit tests, so balances always sum to zero with no float rounding drift, and enforced every permission server-side in Firestore rules rather than client code

File-Lens • Node.js • Express

GitHub | Live

Browser-based storage analyzer, sends only file metadata

- Built and deployed a storage analyzer where users pick a local folder straight from the browser and get instant insights on file size distribution, categories, and modification history
- Separated the Express app from its entry points so the same app runs locally or as a Vercel function, with request validation and pure analyzer functions

ACHIEVEMENTS

- **Processing Foundation OSS Micrograntee '26:** Paid grant to implement the WebGPU instancing API for p5.strands, nominated by project lead Kit Kuksenok from ~120 GSoC applicants
- **CNCF KubeStellar – Admiral Badge:** Awarded for high-volume, high-impact contribution to the KubeStellar project

TECHNICAL SKILLS

Languages: JavaScript, Python, Go

AI/ML & LLMs: NumPy, Pandas, Matplotlib, LLM API Integration, Supervised Learning, Neural Network Fundamentals

Frameworks/Libraries: React, Node.js, Express.js, p5.js

Systems & Automation: Kubernetes, Docker, Linux, Git

Databases & API: Firebase, MongoDB, MySQL, REST APIs

Graphics/GPU: WebGL, WebGPU, GLSL/WGSL